

MargAI — Agentic Road-Damage Survey from Dashcam Video

OpenCV AI Competition 2026 proposal · Team **MargAI** · Himanshu Kumar (solo) · India · a-9724@kmcllu.ac.in · github.com/himanshu748/marg-ai

1. Problem statement and intended impact

Indian municipalities maintain over 6 million km of road with manual, complaint-driven inspection. Potholes are a documented cause of thousands of road deaths per year, yet the data a maintenance engineer needs — *where* the damage is, *how severe*, and *whether it is confirmed* — is collected sporadically and by hand. Dashcams and smartphones already ride along on municipal vehicles, buses and delivery fleets. MargAI turns that ordinary footage into de-duplicated, geo-located, severity-scored damage instances and then lets an agent do what a careful surveyor does: look closer at anything doubtful, ask for a re-survey when frames disagree, and draft prioritised work orders that a human approves before anything is filed. Target users: municipal road-maintenance engineers, smart-city control rooms and fleet operators; beneficiaries: road users.

2. Planned OpenCV 5 image/video analysis

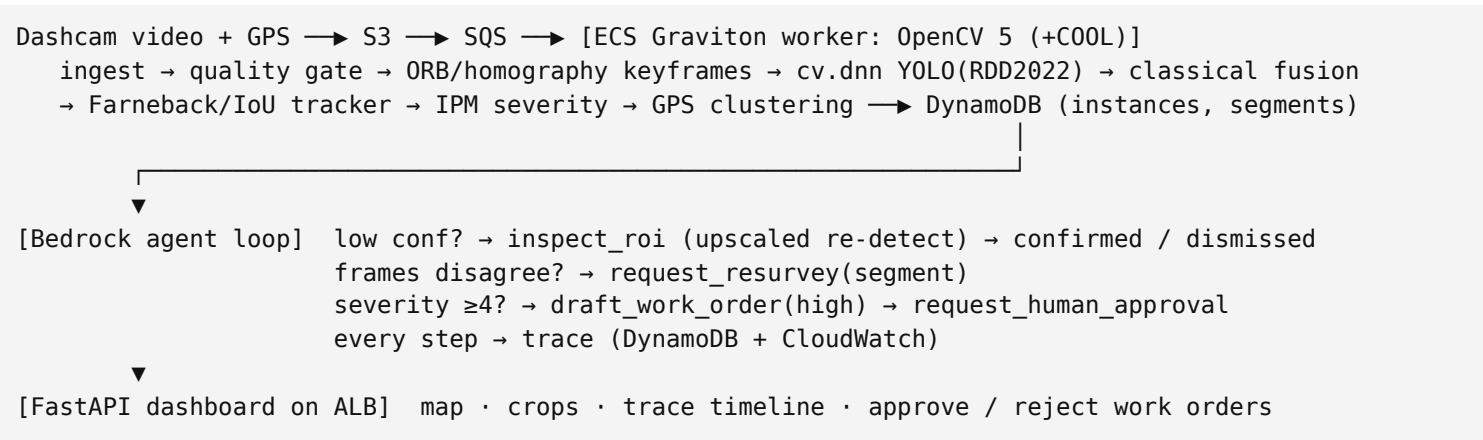
- **Ingest & quality gating** (videoio, imgproc): frame sampling, Laplacian-variance blur rejection, histogram-based exposure rejection.
- **Keyframe selection** (features2d, calib3d): ORB features + RANSAC homography between frames; near-duplicate frames (high inlier ratio) are dropped so the detector runs only on novel views.
- **Damage detection** (dnn): YOLOv8 trained on RDD2022 (classes D00/D10/D20/D40) run entirely through cv : : dnn ONNX inference; we will fine-tune on the RDD2022 India split and report mAP@0.5 per class.
- **Classical support cue** (imgproc): dark-region ratio + Laplacian texture inside pothole candidates fused with DNN confidence to suppress shadow/manhole false positives.
- **Cross-frame instance tracking** (video/optflow): Farneback flow propagates previous boxes; IoU association merges the same defect seen in many frames into one instance with a per-frame confidence history.
- **Severity sizing** (calib3d): inverse-perspective mapping of the road plane to estimate defect area in m²; class + area → severity 1–5.
- **Geo-clustering**: GPS track (GPX/SRT) aligned to frame timestamps; instances clustered into road segments.
- **Active perception for the agent**: the `inspect_roi` tool re-crops and upscales the region and re-runs the detector; the result changes the agent's next action.

3. Planned AWS architecture

Component	Service	Role
Upload & artefacts	Amazon S3	Survey videos, keyframes, crops, agent traces
Job queue	Amazon SQS	Decouples uploads from processing
Vision worker	Amazon ECS on AWS Fargate / EC2 Graviton (arm64)	Runs the OpenCV 5 pipeline; Arm path uses COOL for the core video workload, benchmarked against the x86 pip build (fps, latency, \$/survey)

Agent	Amazon Bedrock (Claude / Nova, Converse API tool-use)	Perception → decision → action loop with tools: inspect_roi, compare_frames, request_resurvey, draft_work_order, request_human_approval
State	Amazon DynamoDB	Surveys, damage instances, work orders, audit trail of every tool call
API & dashboard	Application Load Balancer → FastAPI container	Public web endpoint: map of instances, crops, agent trace timeline, approve/reject
Ops	CloudWatch, IAM least-privilege roles	Logs/metrics, observability of agent decisions

4. Architecture



5. Evaluation method and judge demonstration

- **Detector:** mAP@0.5 per class and latency on the RDD2022 India validation split; comparison before/after fine-tuning.
- **De-duplication:** unique-instance precision/recall against a hand-labelled clip.
- **Agent:** scripted scenarios (low-confidence pothole, severe pothole, frame disagreement, budget exhaustion) scored by an automatic policy auditor: did it inspect before deciding, escalate severe damage, request approval, respect the tool budget? Task-success rate reported for Bedrock and for a deterministic baseline.
- **COOL vs pip OpenCV:** same pipeline, same clip — fps, p95 latency, cost per survey-km on Graviton vs x86.
- **Failure cases** documented: night footage, wet roads, manhole covers, speed-bump confusion.
- **Demo:** live public URL; judges upload a clip (or pick a sample) and watch instances appear on the map and the agent trace unfold, then approve a work order. 5-minute video walks through the architecture and results.

6. Featured paths

We intend to pursue **both** the **Agentic Vision** path (OpenCV output drives subsequent tool calls, with human-in-the-loop approval and full traces) and the **Best Use of COOL** path (core video workload on Graviton via COOL with a reproducible baseline comparison).

7. Team

Himanshu Kumar — B.Tech student, Lucknow, India. Solo builder with a track record of shipped hackathon projects: WeMakeDevs × Cognee (Vegas Amnesia, memory-lifecycle AI game), TxODDS World Cup Hackathon (Matchday Pulse), DevNetwork API + Cloud + AI Hackathon 2026 (LaunchPilot), OpenBMB Build-Small hackathon, an on-device Arm64 FP32-vs-UINT8 benchmarking project (FrameKind), and reinforcement-learning quadruped locomotion trained on a single AMD GPU (Chaal). Comfortable with Python/OpenCV, ONNX inference, FastAPI and AWS deployments.